James Harris

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Results-oriented and innovative Senior Software Engineer with 13 years of experience. Easily communicates complex technical requirements to non-technical stakeholders. Excellent leadership record of leading development teams in enterprise-wide development projects.

Skills

- Software solution building
- Solutions deployment
- Hardware and software installation
- Team reporting

Work History

Senior Software Engineer / Engineering Manager

Crown Consulting, Inc, Pine Brook, NJ *February 2021 - February 2025*

- Developed and optimized high-performance **Java** applications, utilizing core **Java** concepts such as multithreading, collections, and streams to improve efficiency and scalability in enterprise-level applications.
- Designed and implemented RESTful web services using **Java** and **Spring Boot**, ensuring modularity, reusability, and seamless integration with front-end systems and third-party APIs.
- Designed and developed RESTful APIs with **Java** (**Spring Boot**, JAX-RS) for integration with various internal and external systems in both **GCP** and **Azure** environments. Focused on efficient API design, error handling, and security practices, ensuring a seamless user experience and robust backend system.
- Led migration of legacy monolithic applications to **microservices architecture** using **Java** and **Spring Boot**, deployed on **Azure Kubernetes Service** (AKS) and **GCP's Compute Engine**. Enhanced system performance by refactoring existing workflows into decoupled, reusable microservices that improved scalability and maintainability.
- Developed and maintained enterprise-level applications using **Spring Framework** (**Spring Boot**, **Spring MVC**, **Spring Security**), delivering secure, scalable, and high-performance solutions for clients across various domains.
- Designed and optimized **SQL** database schemas, queries, and stored procedures for high-performance data retrieval and manipulation, ensuring seamless integration with **Java** applications. Leveraged tools like **Hibernate** and **JPA** to manage database interactions while maintaining data integrity and optimizing query execution.
- Developed and maintained **NoSQL databases** (such as **MongoDB** and **Cassandra**) for highly scalable and fault-tolerant **Java** applications, implementing efficient data models to handle unstructured data and support rapid data retrieval in real-time applications.
- Led migration of legacy monolithic applications to **microservices architecture** using **Java** and **Spring Boot**, deployed on **Azure Kubernetes Service** (AKS) and **GCP's Compute Engine**. Enhanced system performance by refactoring existing workflows into decoupled, reusable microservices that improved scalability and maintainability.

- Implemented cloud-native **Java** microservices on **GCP** leveraging **Kubernetes** (GKE), ensuring the scalability and reliability of mission-critical applications. Automated deployment pipelines using **CI/CD** practices with **Jenkins** and integrated **Google Cloud Pub/Sub** for real-time event-driven communication between microservices.
- Leveraged **AWS** services, including **EC2**, **S3**, and **Lambda**, to architect cloud-native solutions for deploying and maintaining **Java** applications, ensuring cost-efficient, scalable, and reliable infrastructure.
- Implemented end-to-end automation using **AWS CloudFormation** and **Terraform** for provisioning and managing **AWS** resources, streamlining deployment and enhancing infrastructure as code practices.
- Orchestrated containerized **Java** applications with **Kubernetes**, automating scaling, load balancing, and ensuring high availability of critical services in production environments.
- Led the migration of legacy monolithic applications to **microservices architecture** using **Spring Cloud**, significantly improving system maintainability, fault tolerance, and scaling across distributed environments.
- Containerized **Java microservices** using **Docker**, enabling consistent development, testing, and deployment environments across various platforms, improving system scalability and reducing deployment time.
- Orchestrated containerized **Java** applications with **Kubernetes**, automating scaling, load balancing, and ensuring high availability of critical services in production environments.
- Implemented **CI/CD** pipelines for containerized **Java** applications, using **Jenkins** and **GitLab CI**, to streamline the deployment process and ensure faster time-to-market for new features.
- Integrated **Scala**-based services with distributed systems such as **Kafka** and **Hadoop**, enhancing data throughput and improving overall system reliability and fault tolerance.
- Built asynchronous background tasks in **Python** using **Celery** and **Redis**, enhancing data processing capabilities and reducing response times in large-scale applications.
- Developed **Android** applications with **Kotlin**, utilizing its null safety and concise syntax to improve code quality and reduce bugs, while leveraging **Jetpack** libraries for UI and architecture consistency.
- Designed and implemented **backend services** using **Kotlin** with **Spring Boot**, ensuring clean, scalable, and maintainable code that adheres to **microservice principles**.
- Optimized **React** components for performance, utilizing lazy loading, code-splitting, and memoization techniques to improve the overall user experience, particularly in large-scale applications.
- Leveraged **Python** (**Django**, **Flask**) to build robust backend services, implementing authentication, authorization, and data validation protocols to enhance the security and reliability of applications.
- Implemented **continuous integration/continuous deployment** (**CI/CD**) pipelines using **GitHub Actions** and **Jenkins**, automating deployment of both **Node.js** and **Python**-based **microservices**, ensuring faster delivery cycles.

Senior Software Engineer

BAE Systems, Inc, Totowa, NJ *August 2016 - November 2020*

- **Integrated Spring Security** into Java-based applications, implementing robust authentication and authorization mechanisms, including **OAuth2** and **JWT**, to ensure secure access to RESTful services.
- Designed and implemented **backend services** using **Kotlin** with Spring Boot, ensuring clean, scalable, and maintainable code that adheres to microservice principles.

- Refactored existing Java codebases to **Kotlin**, improving code readability and reducing boilerplate code, which enhanced development speed and reduced technical debt across projects.
- **Integrated PostgreSQL** with Java-based applications, utilizing **JDBC** for efficient data transactions, optimizing queries for performance, and implementing complex relational data models to meet business requirements.
- Designed and maintained highly available and fault-tolerant **PostgreSQL** databases on **AWS RDS**, incorporating automated backups, replication, and disaster recovery solutions for mission-critical applications.
- **Integrated Java microservices** with **GCP** and **Azure** services (e.g., **BigQuery**, **Azure Cosmos DB**) to provide efficient data processing, reporting, and analytics, while ensuring security best practices such as **OAuth 2.0** and **JWT**-based authentication for API access.
- **Integrated relational databases** like **MySQL** and **PostgreSQL** with Java backend applications, utilizing **Spring Data JPA** for efficient data access and ensuring optimized queries for large datasets in a high-load environment.
- Optimized **NoSQL databases** like **Redis** for caching in Java applications, reducing response times and enhancing scalability by leveraging Java clients for efficient key-value storage and retrieval operations.
- Designed and implemented hybrid data models that combined **relational (SQL)** and **non-relational (NoSQL)** databases to meet the needs of different data storage and access requirements in complex Java-based enterprise systems.
- Designed and implemented **RESTful APIs** in **Java** using **Spring Boot**, ensuring efficient communication between front-end and back-end systems while maintaining high performance and scalability for large-scale applications.
- Created containerized Java **microservices** using **Docker** and deployed them on **Azure Kubernetes Service (AKS)** and **Google Kubernetes Engine (GKE)**, implementing effective monitoring and logging practices using **Prometheus** and **Grafana** to ensure application health and performance.
- **Designed and implemented scalable event-driven systems** using **Apache Kafka**, enabling real-time data streaming and processing across microservices architectures, improving system responsiveness and data consistency.
- Developed and optimized **Java-based Kafka consumers** and producers, ensuring fault-tolerant, high-throughput message processing while integrating seamlessly with distributed systems.
- **Integrated Java microservices** with **GCP** and **Azure** services (e.g., **BigQuery**, **Azure Cosmos DB**) to provide efficient data processing, reporting, and analytics, while ensuring security best practices such as **OAuth 2.0** and **JWT**-based authentication for API access.
- Integrated **Python data analysis tools** (**Pandas**, **NumPy**) into backend services, providing data-driven insights through **machine learning models** exposed via APIs built with **Flask** and **FastAPI**.
- Developed and optimized high-performance data processing systems using **Scala** and **Apache Spark**, ensuring efficient data transformations and real-time stream processing for large-scale datasets.
- Collaborated with cross-functional teams to develop and deploy **server-side rendering (SSR)** applications with **React**, improving SEO and loading times for complex web applications.
- Developed and maintained scalable **React applications**, implementing state management with **Redux** and hooks, ensuring dynamic and responsive user interfaces for web-based platforms.

- Designed and implemented **RESTful APIs** using **Node.js (Express)**, integrating them with frontend **React** components to ensure smooth, real-time communication between client and server.
- Developed real-time chat applications and notification systems using **Node.js** and **WebSockets**, ensuring instant messaging and updates between users.
- **Integrated Golang-based microservices** for high-throughput, low-latency data processing in distributed **Java applications**, utilizing concurrency patterns such as **goroutines** and **channels** to handle asynchronous tasks, while maintaining seamless communication between **Java** and **Go** components via **gRPC** and **REST APIs**.

Software Engineer

PGIM, Newark, NJ *March 2014 - August 2016*

- **Developed** and deployed highly scalable **microservices-based** applications using Java (Spring Boot, Spring Cloud) for cloud platforms like GCP and Azure, ensuring seamless integration of distributed services and high availability. Optimized APIs for performance, reducing response times by 25% through asynchronous processing and efficient **caching** strategies.
- **Collaborated** with cross-functional teams to define application requirements, design architecture, and implement **microservices-based solutions** using Java, Spring Boot, and Spring Cloud, enhancing the modularity and scalability of systems.
- **Integrated** relational databases like MySQL and PostgreSQL with Java backend applications, utilizing Spring Data JPA for efficient data access and ensuring optimized queries for large datasets in a high-load environment.
- **Wrote** unit and integration tests using JUnit and Mockito for Java applications, ensuring high code quality and minimizing defects in production environments.
- **Leveraged** Spring Batch to build batch processing applications for high-volume data processing, optimizing job performance and error handling for large datasets.
- **Developed** and deployed highly scalable **microservices-based** applications using Java (Spring Boot, Spring Cloud) for cloud platforms like GCP and Azure, ensuring seamless integration of distributed services and high availability.
- **Developed** responsive, **cross-browser** compatible web applications using HTML5, CSS3, and JavaScript, optimizing user interfaces for mobile and desktop devices, and ensuring seamless integration with back-end services.
- **Created** and maintained microservices with REST APIs in Java (Spring Boot), supporting authentication and data exchange between distributed systems while ensuring API security with **OAuth2** and **JWT** tokens.

Education

Bachelor of Science in Computer Science July 2013

Stockton University, Absecon, NJ